# Global system for mobile communications— what's in store?

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*Global system for mobile communications (GSM) is a grand title. Its aims are to provide complete mobile communications services for its users and network operators around the world. But what does this mean on the ground? Where has GSM come from? Where is it now? And most importantly, where is it heading? This paper sets out to provide some of the answers.*

## 1. Introduction

The original aims of the Groupe Spéciale Mobile of CEPT (European Conference of Posts and Telecommunications Administrations) were to provide a harmonised mobile service across Europe, opening up large markets and thus reducing infrastructure and handset costs. The existing analogue systems in the world provided for the modest needs of the 1980s' user — a high-capacity mobile system with automatic re-tuning to the nearest transmitter ('handover'), good speech quality, basic divert services and voicemail. Yet the markets within Europe both for infrastructure and for handsets were fragmented between the UK Total Access Communications System (TACS), the French Radiocom 2000 system, the German C-Netz system and the Nordic Mobile Telephone (NMT) system. What everyone wanted was a new system which could be universally adopted making a single large market for handsets, base sites and switches.

GSM could have simply copied the best bits of all the first generation services, but there were some basic changes which were desperately needed. Capacity and consistency drivers forced GSM along the digital road. A digital system s much better at coping with multiple handsets operating at the same frequencies on different cells — an interfering signal is simply ignored until the interference level prevents any useful communication and then the line goes silent. This means that the distance between two channels occupying the same frequency space can be reduced for a digital system and hence the capacity of that system is greater than an equivalent analogue system. At the same time, minor interference is completely shrugged off by a digital system and the annoying sporadic rushing noises of analogue systems are eliminated.

## 2. Radio technology

There are many choices for a digital system, but, when the GSM committees had to take a decision, the leading contender was time division multiple access (TDMA), a technology which brings a number of benefits to mobile phone design, making it easier to implement. 'Frequency hopping' (constantly moving the carrier frequency during a phone conversation) was also included to improve performance during 'fading' conditions when the signal path is inadequate to carry the wanted speech. Frequency hopping 'shares' the 'bad' channel between a number of users such that each user perceives only a short gap in speech.

## 3. Security

Security on the analogue systems has been shown to be weak, with continuing headlines of phone tapping using cheap scanners, and copying of phone identities to make fraudulent calls. It was recognised that this problem would increase, and that a leap in security against both eavesdropping and fraud would be needed for the new system. With a digital system the problem is immediately reduced because the use of ordinary voice scanning equipment (originally intended for other more legitimate purposes) would be thwarted. However, it was felt necessary to increase the protection by the following:

- encrypting all speech and signalling,
- using temporary identities,
- employing a strong authentication algorithm.

The speech (and any other user data ) is encrypted using a stream cipher which is applied in the handset and in the base station. This cipher is based on the TDMA ‘frame number’, which only repeats approximately every 3 hours. This prevents the bulk of decryption techniques which rely on pattern following, and would therefore require data from a single call of more than 3 hours duration.

The identification of individuals is protected by the use of temporary identities, so while much of the activity, which has to take place unencrypted (at the start of any communication with the network), can be understood by an eavesdropper, it cannot be related to an individual phone user.

Existing analogue authentication mechanisms are programmed into the phones and thus difficult to change. With GSM it was decided to put the authentication mechanism onto a removable ‘smart’ chip card or Subscriber Identity Module (SIM). This means that the algorithm in use can in principle be altered by changing the SIM. The use of a SIM would also enable each user to keep their personal information, such as phone book entries, separate from the phone itself. Without a SIM inserted, the GSM phone will only allow emergency calls to 112.

All of these mechanisms taken together means an increase of several orders of magnitude in the lengths to which someone would have to go in order to eavesdrop, or more importantly, make off-air copies of a cellular account’s identity for the purpose of making fraudulent calls.

## 4. Roaming

Prior to GSM, roaming was not seen as an essential part of multi-national cellular service. It was included in the North American analogue (AMPS), the UK analogue (TACS) and the Nordic analogue (NMT) systems, but more as an afterthought than by design from the beginning. Complex arrangements have come about, in the United States in particular, where in general you need to register your phone’s identity with the local provider, and the status quo is that roaming is charged per day as well as per call. GSM was designed to provide homogenous service in whichever country the user happens to be. All the user does is switch on the phone as normal, and, provided that service is available and the correct subscription parameters are enabled, everything works just as at home. The automatic transfer of account (subscription) information from home network to roamed network, and billing information from roamed network to home network is a basic part of the GSM philosophy. Charging is also simplified compared with the situation in North America, as the only charges made are for calls, based on the charges which would be paid by a local user of the roamed network plus a declared percentage mark-up. The introduction of the ‘+’ function on the GSM handset was visible proof that roaming was designed-in from the outset; this key allows numbers to be stored in ‘international format’ on the SIM. The actual local international access code is unimportant to the user and is inserted as necessary by the local network.

## 5. ISDN services

In 1982, when GSM started, it was considered that ISDN would be all-pervasive by the time the first GSM networks came on line; this set the scene for introduction into GSM of ISDN-based signalling and ISDN-compatible services. GSM does not have the bandwidth of ISDN, but all the signalling has been provided to allow services to propagate over a GSM link, albeit at a slower rate. As a starting point, it was intended to provide all of the ISDN supplementary services (call hold, call waiting, caller identity, etc), plus data services and group 3 facsimile. Not all of these could be achieved in the first GSM phase, but the basic ones were included.

## 6. Short message service

It was thought that all of the above facilities might not quite be enough to encourage everyone to move away from the analogue services and on to GSM. At the same time the growth of paging services had led to many users carrying both a radiopager and a mobile phone. This brought about the Short Message Service, which allows 160 character messages to be sent to or from a GSM phone. Messages are acknowledged, and if they cannot be delivered because a phone is switched off or out of coverage, they are stored by the network and retransmitted the next time the phone is available.

## 7. GSM phases

The first GSM systems and handsets to be introduced were referred to as phase 1. It was originally intended that the design work for GSM would be completed by 1991 and that the systems would start straight away. However, as that date approached it became clear that the standards had been too ambitious and that not everything could be finalised in time. Therefore phase 1 had to be ‘frozen’ to prevent further change and allow manufacturers to implement the specification. The idea was that phase 2 would then follow to complete the original aims. As anything which was not close to completion and not regarded as essential was delayed until phase 2, services like call waiting and caller identity were delayed. The manufacturers also decided not to implement everything in phase 1, because the perceived market need only included a subset of the facilities. Consequently, a number of services (such as SMS, data and fax) have been called ‘phase 2’ by the media when, in fact, they were included in the phase 1 standard.

Phase 2 was to complete the original picture. However, the market for cellular services was changing very rapidly and what originally exceeded the market needs was now short of facilities in several areas. Phase 2 therefore included some additional facilities to catch up with the market requirements. The work started in parallel with completion of phase 1, but it was soon realised that phase 1 had not been designed in a completely forward-compatible manner. The original specifications had not included sections on how a mobile or a network element should behave if a received signalling message fell outside the specification. This significant omission immediately meant that the introduction of new signalling messages for phase 2 services would result in unpredictable behaviour in some cases. Therefore the design engineers had to grasp the complexities of backward compatibility and decide to introduce a number of key changes which in some cases meant changes to existing implementations. This was needed to provide a stable platform for future enhancement.

Towards the end of phase 2, it was realised that the GSM market requirements were unlikely ever to stop moving, and that further standards work would be required for some time to come. Rather than introduce a new phase 3, it was felt that all future standards work should be based on phase 2, and a collection of enhancements should be made public from time to time — this is called phase 2+.

## 8. Building on the standards

The standards provided a basis for launching mobile networks and this has proven very successful. Earlier systems provided an open interface only in the radio link between the base station and handset; however, the GSM system has open interfaces (see Fig 1) between many of the network components as well:

- between radio base station and base station controller,
- between base station controller and local mobile switching centre,
- between local mobile switching centre and gateway mobile switching centre,
- between all mobile switching centres and subscriber database,
- between all mobile switching centres and equipment database,
- between GSM networks for connecting calls and short messaging, between GSM networks for exchange of billing information.

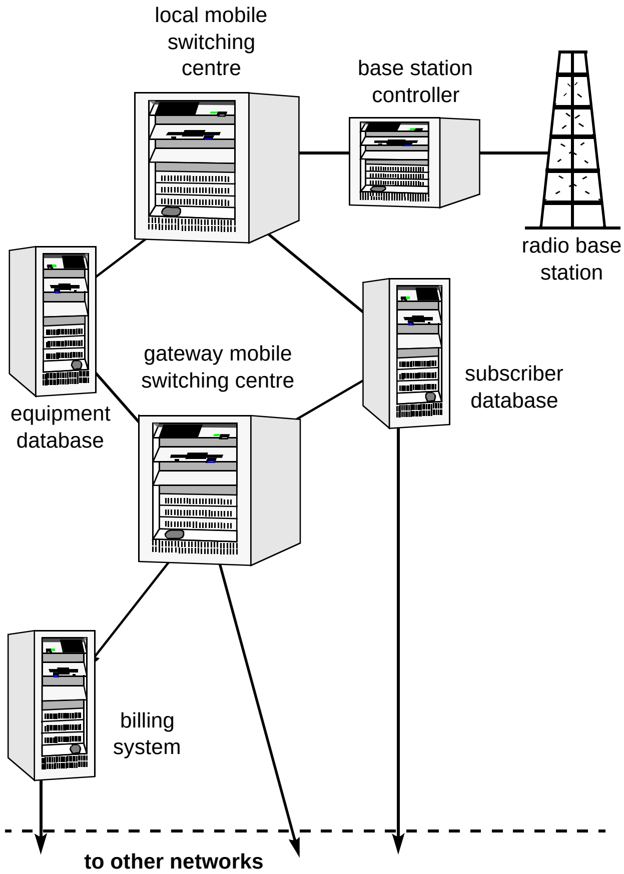


Fig 1 Open interfaces.

In providing these very open interfaces, GSM has allowed competitive procurement of nearly all network components. This has led to the most open market in the world for a cellular system, and there are firm drivers for manufacturers to ensure that their equipment has the best price/performance, as operators are no longer ‘tied in’ to single manufacturers for the whole network.

These drivers have resulted in a dramatic reduction in prices and also a rapid expansion of GSM service across the globe. This size of market in turn has driven the manufacturers to provide even more advanced equipment, for example, very small base stations now exist which just cover a single building.

It is not only the network components which have matured very quickly — the size of the GSM market is forcing the handset prices down and new features are coming along at a rapid rate; however, many of these features are only a part of the total service and there is still much to be improved in the networks and handsets to achieve the ultimate service ease-of-use. This is particularly true of SMS, e.g. where the basic service of delivering text to and from a phone was provided in some very early phones, but until people started to use it for real purposes it was not possible to design a good user interface. With data services, too, manufacturers have tended to provide services based around PCMCIA or PC cards — this makes life difficult when you are trying to use a computer without a PC card slot.

In other areas, too, experience of running a GSM network has been the only way to discover incompatibilities. With ISDN signalling, for example, it is possible to describe the exact nature of a call, whether it is data (and which type), facsimile or speech. Many facsimile machines are connected to PBXs which support ISDN signalling; however, if an incoming call indicates ‘facsimile’ instead of ‘speech’ or ‘PSTN audio’, the call is sometimes mistakenly rejected by the PBX because of an incomplete signalling implementation. This situation never occurs elsewhere because a normal facsimile machine is incapable of indicating ‘facsimile’ in the outgoing signalling for the call. Consequently, enhancements are needed not just to the GSM network, but also to some non-standard *de facto* fixed network components.

## 9. What’s next?

Though there is much to be done in developing, building and marketing the existing standards, GSM is not standing still. A number of phase 2+ developments are under way. They fall into a number of natural groups which are discussed in more detail in the following sections.

### *9.1 Customisation of services*

In some respects the standardisation of GSM has been too successful. With the very large markets, the manufacturers of equipment are becoming more and more reluctant to produce special versions for individual companies because they can make more money by advancing the barriers and breaking into the newest versions of the standards. The original intentions of standardisation in leaving open every avenue for individual operator innovation have thus been stifled by a lack of power to innovate.

At the same time, the ‘intelligent network’ concepts have become more crystallised, with operators wishing to be able to introduce new services rapidly and on a small scale if necessary. Although GSM does embrace some of the basic intelligent network concepts, there are many areas where enhancements are needed.

Calls are currently not always routed in the optimum way. For example, two GSM users roaming abroad needing to call back to the UK to speak to one another, may result in two expensive international calls to connect two people who could be in the same country or even city. Incoming calls which cannot be connected because the phone is switched off are transferred back to the home country, again resulting in two international calls. Such inefficiencies are resolved by GSM ‘optimal routeing’, which adds intelligence to the switch fabric of the GSM network and re-routes calls according to the most efficient scenario.

This is only the start of the more intelligent GSM networks. New features are now being developed under the ‘customised application for mobile networks enhanced logic’ or CAMEL banner. PLMN operators who implement the CAMEL features will have the capability to provide operator specific services (OSS) that are new and unique. The CAMEL features complement and add to the ‘standard’ GSM supplementary services as implemented by all GSM operators. Operator-specific services can be extended to serve customers roaming to any other GSM network which supports the CAMEL features.

Figures 2 and 3 show the network architecture for call handling and call forwarding before and after the introduction of CAMEL, which allows a much greater control of outgoing calls by the home networks, as described below.

CAMEL requires exchange of customer-specific information between GSM networks. Before exchanging information it is necessary to identify the CAMEL service environment (CSE) responsible for executing the customer’s service logic. This information is stored by the HLR in the CAMEL subscriber information (CSI). The CSI indicates the subscriber that requires CAMEL support and identifies the CSE. The gateway MSC (GMSC) in the interrogating network (IPLMN) obtains the CSI from the HLR when a new mobile terminating (MT) call is initiated. The VLR (and hence visited MSC) in the visited network (VPLMN) obtains the CSI from the HLR as part of a location update, along with other subscriber service information.

The VPLMN and IPLMN could be the same GSM network as a calling or called subscriber’s home network (HPLMN), or separate GSM networks, in any combination, (provided the separate networks support the CAMEL feature).

The information exchanged between CSE and MSC is defined by the following procedures.

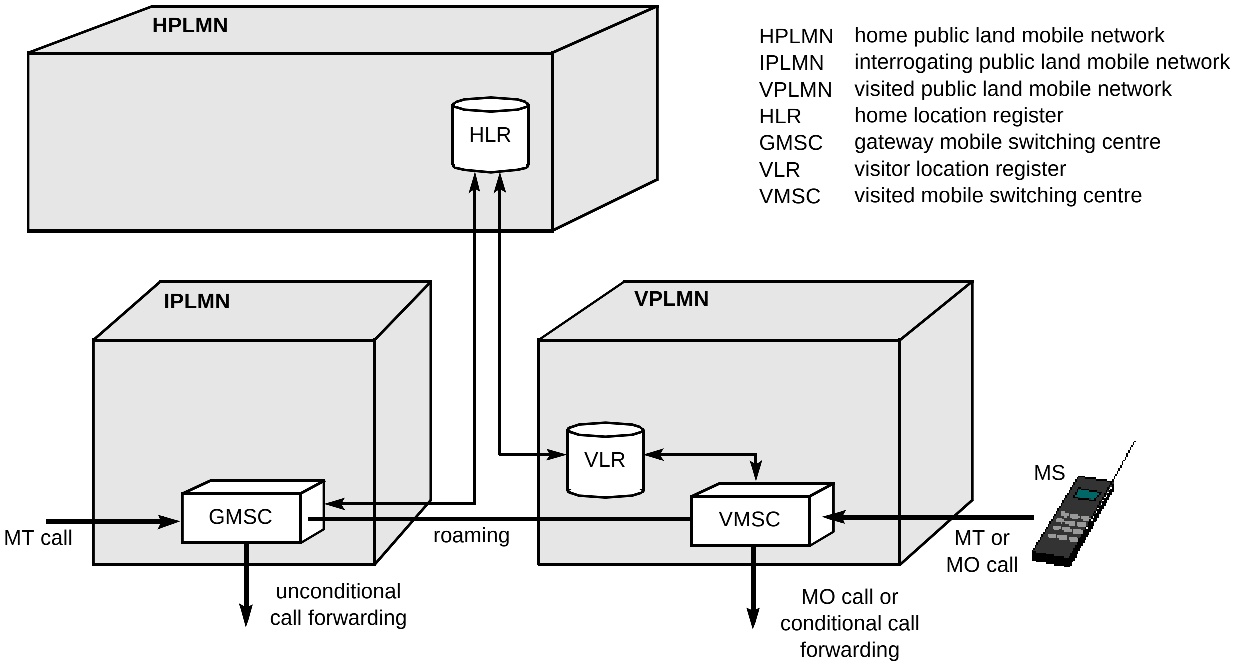


Fig 2 GSM network architecture — before CAMEL.

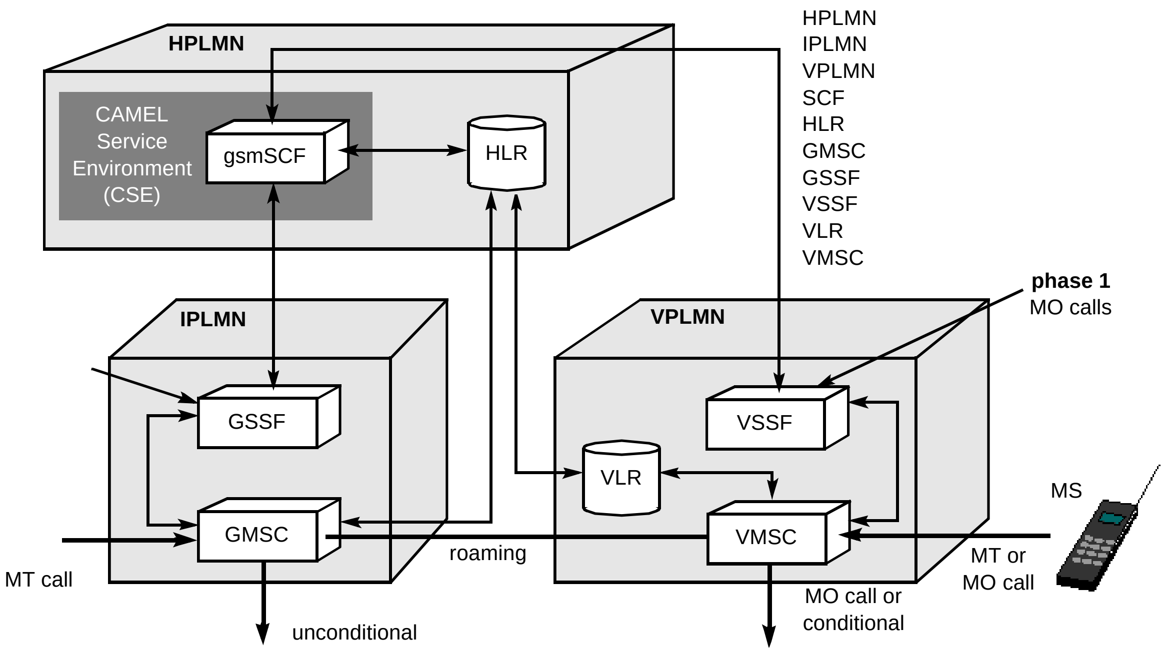


Fig 3 GSM network architecture — after CAMEL.

- The MO *call set-up request* procedure allows the CSE to modify how call set-up is carried out when any number is dialled or a specific number is dialled.

- The MT *incoming call request* procedure allows the CSE to modify how call set-up is carried out when the called subscriber CSI indicates CAMEL support is required.

- The *call set-up monitoring until failure* procedure is applied before connection is established. It allows the CSE to modify how a call is progressed when the call is not answered by the called party, the calling party clears, or a network error occurs.

- The *called party connection* procedure allows the CSE to influence call handling, such as charging, when the called party answers and the connection is established.

- The *call terminating* procedure allows the CSE to modify how a call is cleared down when an established call is terminated by the calling party, called party, or for any other reason.
- The *change of parameters of established call* procedure allows the CSE to modify how a call is handled when a tariff rate change is necessary.
- The *any time interrogation* procedure allows the CSE to request information about a subscriber's status (idle, busy, detached) and location.

The *any time interrogation* procedure is very important for mobile customers. Mobile status parameters, such as busy, and mobility information, such as location, can be remotely monitored so that important events that impact on service functionality can be detected and used to influence service logic.

When a CAMEL subscriber roams to a VPLMN that does not support CAMEL the HPLMN determines the service offered. Operator specific services may be partially supported or not supported at all.

### *9.2 Enhancements to data services*

The GSM phase 2 data services provide a very good robust channel for transmission and reception of data on the move. However, they require the use of a whole traffic channel while the data context is established at the mobile end and the fixed end. The phase 2 services are particularly suited to a mode of working where the user logs on, uploads data, downloads data and then exits. Many data applications, however, require the transmission and reception of data packets from time to time via a semi-permanent virtual link. Such applications can work over a phase 2 data call, but this has an impact both on cost and capacity requirements. From the network operator's perspective, the data user is taking up capacity which is not needed, and from the user's perspective, the cost is in most cases prohibitive. The generalised packet radio service (GPRS) overcomes these shortcomings by allowing the establishment of virtual circuits which use physical resources only when data is required to be transmitted or received.

It is also perceived that the highest data rate over GSM (9.6 kbit/s) is quite slow by modern fixed network modem standards. The high-speed circuit-switched data (HSCSD) work item is addressing this by specifying the use of multiple time-slots for a single call. Each GSM radio carrier can support up to eight simultaneous calls, each occupying a single time-slot. With HSCSD each call can be allocated up to eight 'time-slots', effectively using as much bandwidth as eight calls. In this way data rates up to 80 kbit/s (or potentially over 300 kbit/s with data compression) could be supported — however, this is not simple to achieve. As the basic GSM handset relies on the 'unused' time-slots for thermal management, allowing the transmitter circuits to cool, a technology advance is required to avoid overheating when transmitting all the time. Additionally, the radio synthesiser, which tunes alternately into the receive frequency and the transmit frequency, will need to be duplicated as transmission and reception must occur in the same time frame.

Even when these problems have been overcome, the question remains as to how the mobile market will react to very expensive high-bandwidth calls. For mobile video and multimedia applications it is obvious that higher bandwidth is required. Yet is it conceivable that the user will be prepared to pay four or even eight times the cost of a voice call for a video call? Such questions will need to be resolved in the cellular market-place before operators can decide to invest in these advances.

### *9.3 Group calling*

Outsourcing is a very popular trend these days and the mobile radio market is no exception. Traditionally, the utility companies and emergency services have rolled out and maintained their own private mobile radio (PMR) networks, but they are now coming to the point where they need to upgrade their systems and are considering carefully whether to attempt to outsource all of their mobile communications to the cellular operators. In addition to this, new spectrum has been set aside for the use of the European Railways networks which would benefit from the large-scale manufacture and technical innovation inherent in GSM. The phase 2 GSM standard, however, cannot meet the needs of the PMR market in the following three key areas, and these are being addressed in phase 2+.

- Group calling — is the ability to hold a conversation between multiple parties on a single radio channel. This allows anyone in a given area (with appropriate authorisation) to listen in and contribute to a discussion. In an emergency this is especially useful, as setting up and controlling a traditional GSM multi-party call is neither fast nor simple.
- Priority calls — one of a number of priority levels is ascribed to each call, and calls with a higher priority are given a higher level of service. This means, for example, that emergency services can be assured a channel even if the cell is busy.
- Pre-emption — enhances the priority mechanism by allowing a high-priority call request to disconnect an existing call with lower priority; this means that the radio resources can be used to service the most important calls in a very effective way.

### *9.4 Advanced supplementary services*

As GSM networks become more integrated with business networks and in particular the private exchanges which serve corporate sites, so the demand for an equality of service availability will increase. Calls from one fixed ‘PBX’ connection to another use only a short code, so if a colleague happens to be away from the office using a mobile phone, why should it be necessary to use a long complicated number? The new GSM service ‘support of private numbering plan’ will allow a GSM user to call and be called as if the GSM handset were a fixed phone on the private exchange. For the GSM user and office-based users this makes it much easier to keep in touch.

One of the most used ‘PBX’ facilities is the ‘ring back when free’ or ‘call completion to busy subscriber’ (CCBS) service. On a private network such facilties are ubiquitous and easy to implement because the calling and called terminals are always attached to the same point in the network, and there are no charges to be billed. On mobile networks, however, users may move about freely so it is not guaranteed that the same switch site will be serving the mobile when the ‘free’ indicator is returned. Indeed, the mobile may even be switched off at this time. This means significant enhancement to the network signalling to permit the ‘free’ indicator to be stored at the appropriate place and forwarded to the original caller when they are next available. There are also billing considerations to be resolved. In the GSM case it has been decided that the user may be charged both for subscription to the CCBS service and for each use of the CCBS service, although network operators may choose to waive these charges as the increased network usage may well outweigh any gain in subscription/activation minus the cost of charging. Naturally, the original caller pays for the duration of the completed call. The signalling is arranged such that the second (post-alert) call is basically the same as a new manually invoked call.

Another well-used ‘PBX’ facility is call transfer, which in GSM is called explicit call transfer. This allows a GSM user to transfer an incoming call to another number and then drop out of the call. In principle the charges are the same as for a three-party call (original caller pays for one call and transferring user pays for the call to the third party), but in this case the transferring user has no control of that call’s subsequent duration. Concerns over this and the potential for fraud have resulted in very detailed studies of all the possible scenarios.

Multiple subscriber profiles will allow a user to have more than one telephone number, so that if required certain incoming calls can be diverted elsewhere. For example, at the weekend a business/personal user might require business calls to be diverted to voicemail but personal calls to come direct to the handset. The service will also allow calls from the GSM handset to be sorted into different billing groups, so that, for example, any personal calls arrive on a different bill from business calls.

Phase 1 included an advanced capability called ‘unstructured supplementary service data’. The intention was to allow phase 1 handsets to support some services which were not defined in phase 1. On receiving a request from a user, the handset normally translates the request into signalling appropriate to the service, for example to enable Call Waiting, the user enters *43#SEND. Because Call Waiting was not completed in phase 1, a phase 1 handset will send a ‘USSD’ message to the network containing ‘*43#’. With a phase 2 phone, however, the code is translated into a ‘REGISTER’ signalling message containing the signalling code for Call Waiting. This concept has now been expanded to enable operators to offer services based on ‘operator-specific’ codes and text responses from the network. An example might be to request the current billing period with, e.g. *#100# and the network response would be ‘PEAK’ or ‘CHEAP’. Further extensions will allow the network to send unsolicited USSD messages to a mobile.

### *9.5 Short message enhancements*

The GSM short message service was completed in phase 1, but there are many facilities which have been added in phase 2, such as the ability to address a message to an alphanumeric tag, rather than a less friendly number, or the ability to send delivery receipts to a mobile. These were thought of as the services were coming towards launch by the various GSM operators. As short message services are widely available from virtually all operators, some of the experiences of commercial availability are being brought to bear on the standard. The UK PCN operators included a non-standard indicator in all their phones which tells the user that there is voicemail waiting. This has now been transferred to the GSM standard in a slightly different way, but mobile manufacturers can now implement four different indicators for voicemail, faxmail, e-mail and a yet-to-be-defined alert. Network operators can then start to use these indicators when mobiles which support these features become available.

GSM was conceived as a Europe-wide system and the character set for SMS reflects this; however, GSM has also had significant success in other markets, and there is now a need to be able to send non-European characters (Chinese, Japanese, Arabic, Cyrillic, etc). A number of attempts have been made to introduce a variety of schemes into the standard to support these characters, but all were flawed by over-complexity of implementation until the character set defined by UCS/2 was proposed. This defines all characters which are required for GSM worldwide, but uses 16 bits per character instead of 7 bits per character. At first sight, this reduction from 160 character messages to 70 character messages looks rather a significant loss, but the far eastern markets can send more data with a 70-character image-based message than can the Europeans with a 160-character text-based one! The main aspects to be resolved are the introduction of a compression mechanism to allow more text to be transferred and interworking between older phones which do not support UCS/2 and newer phones which do.

Linking multiple SMS messages together was also felt to be important. Although non-standard mechanisms for sending text which cannot fit into a single message are available (e.g. by putting some textual information in the message about the message sequence), a standard method allows a message to be sent to an arbitrary destination in the correct format. So the standard now allows up to 255 messages to be 'combined' into a 'macromessage'. This mechanism removes some of the text, but still allows more than 38 000 characters in a macromessage.

### *9.6 SIM enhancements*

The original concept of the SIM as a mechanism to contain all the user's personal information has been tarnished a little by the rise in the use of the small version to keep the handset size down. Users are now unlikely to move SIMs around between phones and the need to keep everything on the SIM is not as great as originally thought. However, the need for network control of the user's phone and availability of features was considered to be restricted to the HLR and the idea of 'updating' the SIM over the air was not originally thought to be necessary. The UK PCN operators changed that by introducing a non-standard 'SIM update' mechanism which enabled them to bar or enable particular phone features on a user-by-user basis. It also made a link between the administrative systems controlling the HLR and the user, so that when something changed within the HLR the user would see 'SIM update' on their screen even if nothing had actually been changed on the SIM. This 'comfort factor' and the ability to control access to various services 'over the air' was felt to be very important. Phase 2+ therefore includes the ability for an SMS message to contain data specifically for programming a SIM; this is automatically transferred by the phone to the SIM and a response may be sent back to the SMS Service Centre to confirm that everything has been updated successfully. Of course, the implementation of these features needs to be carefully engineered to avoid the possibility of users re-programming their own SIMs.

## 10. Conclusions

With all of these features, GSM Phase 2+ encompasses virtually every need perceived for the mobile customer, from high-speed data and group calling to intelligent call re-routeing and projection of home network ser-vices into whichever network you happen to be.

GSM has become much more than a 'second generation' cellular system. At the start it was envisaged that GSM would need some additional features to attract subscribers from their existing phones, but it has subsequently been seen that even the most basic services are important to many. So those extra features which were added to make GSM attractive to the basic user are a bonus. The good news is that this has built a large market for GSM, but the challenge facing the telecommunications industry is that after three years of GSM service many of the 'extra' features are now perceived as basic and people are questioning why all features are not available on and across all networks, both fixed and mobile.

With all these developments GSM is becoming the leading telecommunications system, encompassing fixed network services, PMR services, packet radio services and leading-edge intelligent network services. Before long the only technical advantage of the fixed network will be the bandwidth it can support.

Kevin Holley graduated from Bristol University in 1985 with a BSc in Physics. He joined BT Laboratories that year and he began working on off-air analysis of analogue cellular signalling. His involvement in the global system for mobile communications (GSM) standards work began in 1988 with a study of the reliability of the proposed signalling mechanism. Later in 1988 he began work on the short message service (SMS) and has been closely involved with this ever since. In 1990 he became chairman of the ETSI subgroup developing the SMS standard. In 1989 he was seconded to Cellnet to work on GSM technology development. He returned to BT Laboratories in 1992.

He is currently working on linking value added services with SMS, and further developments in GSM technology for data and value added services.